Home Solar Panel Maintenance Guide

Solar panels are durable, but regular maintenance keeps them working at peak efficiency. A typical residential system of 5 kW produces between 6,000 and 7,500 kWh per year depending on location and orientation.

Dust, pollen and bird droppings can reduce output by 5 to 25 percent. In dry and dusty regions, clean the panels every three months; in rainy climates, twice a year is usually enough.

Cleaning procedure and safety

Always switch off the system at the inverter before cleaning. Use a soft brush or sponge with lukewarm water and a small amount of mild soap. Never use high-pressure washers, because they can damage the seals and the anti-reflective coating.

Clean early in the morning or late in the evening, when the panels are cool. Spraying cold water on hot glass may cause thermal shock and micro-cracks. If the panels are on a steep roof, hire a professional who uses fall protection.

Inspection, inverter and warranty

Inspect the panels visually every six months for cracks, discoloration, loose wiring and corrosion on the mounting rails. Check the inverter display monthly: a green light normally means the system is operating correctly, while a red or orange light indicates a fault.

String inverters usually last 10 to 15 years and are the component most likely to need replacement. Most panel manufacturers offer a 25-year performance warranty that guarantees at least 80 percent of the original output.